Operation of Brushless DC Motor to Drive the Electric Vehicle

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***Abstract*— This paper is to select the Brushless DC Motor size to be suitable to drive the electric vehicle. This selection is corresponding to the specifications of the electric vehicle. A comprehensive investigation of the brushless DC motor as well as the electric vehicle has been accomplished. The vehicle dynamics are also illustrated. And then A detailed method for the sizing the major components of the electric vehicle is presented. Eight switches bridges converter is used to drive the brushless DC motor. The motor characteristics are computed taking into account the effect of the chosen control circuit on the motor currents.**

Keywords*— Brushless DC Motor (BLDCM), Electric Vehicle (EV), Motor Rating, Motor Performance.

I. Introduction

Environmental and economic issues provide a Strong impulse to develop clean, efficient, and sustainable vehicles for urban transportation. Cars are an integral part of our daily lives, However, conventional exhaust emissions from conventional combustion engines are responsible for the main source of urban pollution that causes global warming. [1]. pollution problems increased yearly as a result of the increasing number of automobiles. High-efficiency electric motors and controllers provide an electric vehicle (EV), which used an alternative energy source provide the means for a clean, efficient, and environmentally friendly urban transportation system. Electric vehicles having the potential to curb the pollution problem in an efficient way. Therefore, EVs are the only emissions-free vehicles [2,3,4]. There are four major types of electrical motor drives used for EVs, Brushed DC Motor drives, induction motor drives, Switched Reluctance Motor drives and Brushless DC Motor (BLDCM) drives. BLDCMs are extremely appropriate for EVs to their high power densities, speed-torque characteristics, high efficiency, large speed ranges and maintenance free. Compared with a brushed DC motor. The BLDCM is more reliable than the DC motor, as it depends on electric commutator instead of a mechanical commutator. High efficiency realized in BLDCM, as a reason of the rotor's magnetic flux produced by the magnets installed on the rotor [2]. Brushless construction lead to reduced motor size, control be possible, high reliability and maintenance-free operation. Torque to motor size ratio is high, Thermal overload & under load protection is provided [5-9,14].

II. Modeling of the Mechanical Modeling

The mechanical coupling between motor and wheels is shown in Fig. 1. Electric vehicles are designed without a gear. But, the gearbox permits the electric motor to operate at extremely higher speeds for given vehicles speeds. Then the motor size is minimized because of the low torque demand at higher speeds. [2].

$$m = \left(\frac{Wm\ rwh}{Vveh}\right) \quad (1)$$

The developed torque at elctric vehicle shaft computed from the next equation:

$$T_{d_wh} = m\left(T_e - (b_g * W_m)\right) \quad (2)$$

T_{d_wh}: is the developed torque at the shaft of the wheel
T_e: is motor developed torque
W_m: is the motor speed (rad /s)
V_{veh}: is the vehicle speed (m /s)
b_g : friction coefficient of the gearbox
r_{wh} is the radius of the wheel

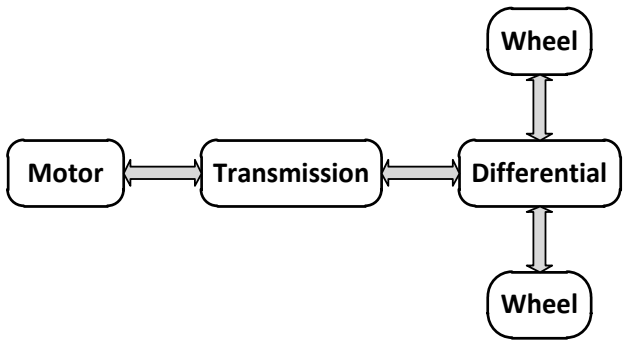

Fig. 1 The mechanical coupling between motor and wheels

A. Propulsion Power

The desired power rating of the electric motor or the power required from the combination of electric motor and internal combustion engine (i.e., the propulsion unit) is studied from the above equations based on the system requirements of starting acceleration, the rating of the vehicle and maximum velocity, and vehicle gradeability.

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III. Mathematical Modeling of BLDCM

BLDCMs are permanent magnet motors where the function of commutator and brushes were replaced by solid-state switches. BLDCMs configure of single-phase, two-phase and three-phase. The three-phase motors are the most widely used. Due to the surface permanent magnet of the motor, it produces a trapezoidal back electromotive force (EMF). The motor current generates a pulsating torque [8,9,10].

Eight switches bridge converter was proposed to achieve electrical separation between phases and increase reliability of the system [13]. As shown in Fig. 3 there are six states govern the motor motion by switching the eight electronic switches shown in Fig. 2 and according to TABLE I.

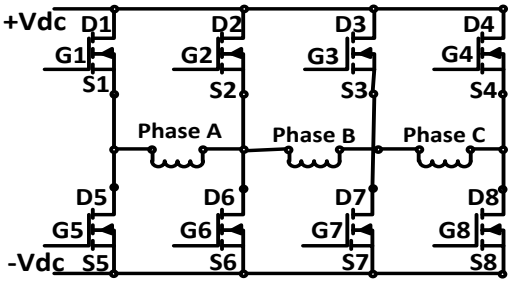

Fig. 2 Eight Switches Bridge Converter

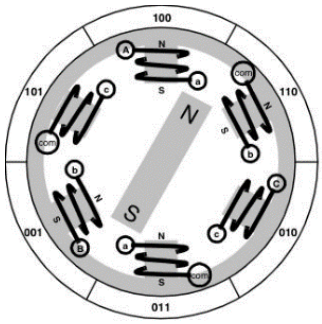

Fig. (3) Brushless DC machine internal diagram

TABLE I.
SWITCHING SEQUENCE OF EIGHT SWITCHES BRIDGE CONVERTER

Switches	Rotor position (degree)					
	0:60	60:120	120:180	180:240	240:300	300:360
Q1	ON	OFF	OFF	OFF	OFF	OFF
Q2	OFF	ON	OFF	ON	OFF	OFF
Q3	OFF	OFF	ON	OFF	ON	OFF
Q4	OFF	OFF	OFF	OFF	OFF	ON
Q5	OFF	OFF	OFF	ON	OFF	OFF
Q6	ON	OFF	OFF	OFF	ON	OFF
Q7	OFF	ON	OFF	OFF	OFF	ON
Q8	OFF	OFF	ON	OFF	OFF	OFF
Currents Direction	+A	+B	+C	-A	-B	-C

A. *Analysis of BLDCM*

The stator phases voltages are:

$$Va = Ia\,R + L\frac{dIa}{dt} + Ea \quad (3)$$

$$Vb = Ib\,R + L\frac{dIb}{dt} + Eb \quad (4)$$

$$Vc = Ic\,R + L\frac{dIc}{dt} + Ec \quad (5)$$

B. *Trapezoidal Back-emf:-*

Due to the surface mounted permanent magnet in the rotor of BLDCM, an e.m.f. is produced in each stator phase winding with rotor rotation. The back-EMF polarity is in opposite direction to the applied voltage per phase. Back-EMF (E) rely on three factors [8]:

- Angular velocity of the rotor
- Magnetic field generated by rotor magnets
- The number of turns in the stator phase windings

$$E = Ke * We \quad (6)$$

where Ke is the back EMF constant
W_e is the rotor speed

Then the instantaneous values of the back-emf of three phases as follow:

$$Ea = Ke * We * Fa(\theta) \quad (7)$$

$$Eb = Ke * We * Fb(\theta) \quad (8)$$

$$Ec = Ke * We * Fc(\theta) \quad (9)$$

To select the BLDCM as an application of the EV, it is important to decide the appropriate scope of the BLDCM parameters. This is accomplished by computing the dynamic acceleration from equation (2) coupled with the electrical equations (3), (4) and (5).

IV. POWER RATING SELECTION OF THE TRACTION MOTOR

According to acceleration performance the power rating of the BLDCM is computed as follow [2,9,15]:

$$P_{BLDCM} = \left(\frac{k_m\, M_{veh}\, V_f^2}{(2\, t_a)}\right) + \left(\frac{2\, C_O\, M_{veh}\, g\, V_f}{3}\right) + \left(\frac{\rho\, C_D\, A_f\, V_f^3}{5}\right) \quad (10)$$

Where:

$\left(\frac{k_m\, M_{veh}\, V_f^2}{(2\, t_a)}\right)$ represents the power used to accelerate the vehicle mass,

$\left(\frac{2\, C_O\, M_{veh}\, g\, V_f}{3}\right)$ represent average power for overcoming the tire rolling resistance and,

$\left(\frac{\rho\, C_D\, A_f\, V_f^3}{5}\right)$ represent the average power of aerodynamic drag.

Then the acceleration time (t_a) is computed from the next equation :-

$$t_a = \frac{k_m\, M_{veh}\, V_f^2}{(2\, P_a)} \quad (11)$$

Where, P_a is the tractive power through vehicle speed from zero to final speed and V_f is the final speed of the vehicle

Then, the acceleration distance (d_a) is computed as follow: -

$$d_a = V_f \int_0^{t_a} \sqrt{\frac{t}{t_a}}\, dt = \frac{2}{3}\, V_f\, t_a \quad (12)$$

The power rating of the electric drive motor in EV is determined by vehicle acceleration requirements. For the vehicle under consideration, Appendix (a, b), by substituting into equation (10), the rated motor power is drawn against the acceleration time as shown in Fig.4. the rated power of the traction motor obtained as 35 kW for the specified

acceleration time of 11.75 seconds from zero to 100 km/h. Motor performance should be checked by vehicle acceleration requirements. Firstly, acceleration time and distance are computed. By substituting in equations (10), (11) and (12) the results of acceleration time and acceleration distance is 9.54 second and 197.5 meters respectively. That is which needed to the vehicle to accelerate from 0 to 100Km\h.

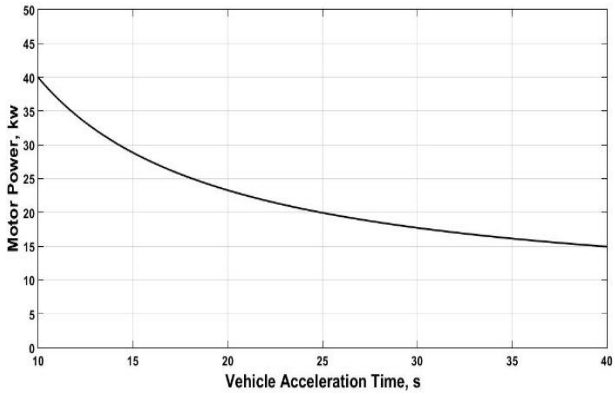

Fig. 4 Motor rated power versus the time needed for accelerate the vehicle from 0 to 100 km/h.

A. *Check The Sizing Of BLDCM In EV Drivetrain*

Using the data of the BLDCM and EV given in appendix (a,b,c), the following characteristics is obtained. Motor developed torque and load torque is obtained against the vehicle speed by varying the motor terminal voltage from (40: 280 v).The steady state operating points are obtained from the intersection between developed torque and load torque. The developed torque of brushless DC motor and load torque is drawn from equations (13) , (16) as shown in Fig. 5.

$$Te = \frac{(Ea*Ia+Eb*Ib+Ec*Ic)}{w_r} \quad (13)$$

$$F_{RL} = C_o M_{veh} g + 0.5\rho C_D A_f V_{veh}{}^2 \quad (14)$$

$$T_{wh} = F_{RL}\, r_{wh} \quad (15)$$

$$T_L = \frac{T_{wh}}{m} \quad (16)$$

Where

F_{RL} is the road load force

T_{wh} is the load torque at the shaft

T_L is the load torque of the motor

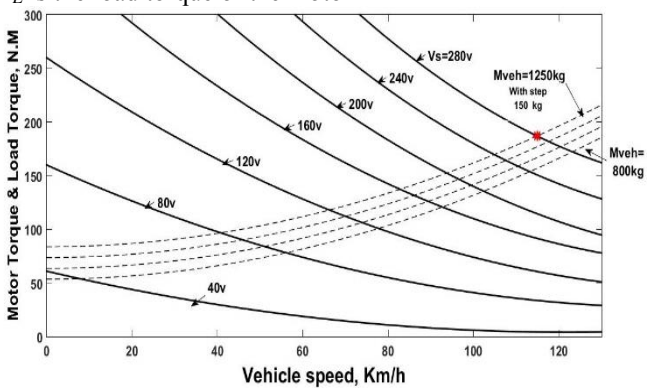

Fig. 5. Motor developed torque and load torque versus vehicle steady-state speed at different values of supply voltages and vehicle mass respectively

Then the motor terminal voltage plotted versus the vehicle steady-state speed at various values of vehicle mass as depected in Fig. 6. It is noticed that vehicle speed increases when voltage increased with large changes. Also, vehicle speed decreases when vehicle mass increases but with small changes.

Using the steady-state vehicle speed, which obtained from Fig.6 at various values of motor supply voltage and vehicle mass, into equation (17) the motor speed is obtained. Then the motor output power is drawn versus the vehicle speed at different values of vehicle mass from equations (13,17,18) as shown in Fig.7.

$$Wr\ (electerical) = Wm\ \ (mechanical) * P\ (no.\ of\ pair\ poles) \quad (17)$$

$$P_{out} = T_e\, w_r \quad (18)$$

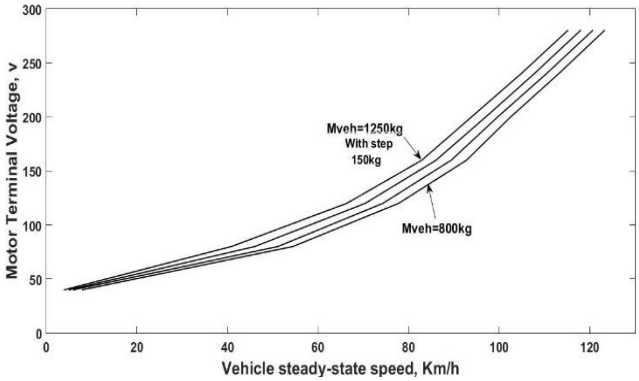

Fig.6 Motor teminal voltage versus vehcile steady-state speed at different values of vehicle mass

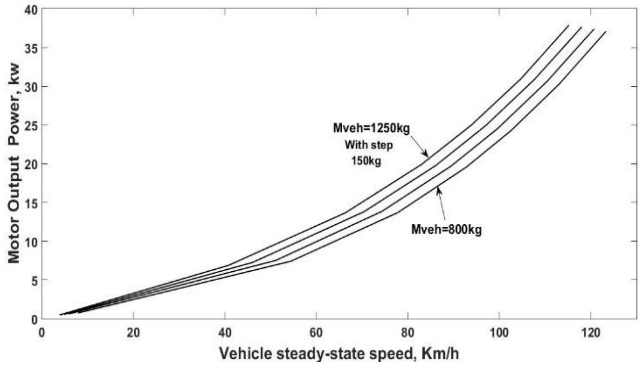

Fig.7 Motor output power versus vehcile steady-state speed at different values of vehicle mass

From equation (19) the efficiency of the brushless DC motor is drawn against vehicle steady-state speed at different values of vehicle mass as illustrated in Fig. 8. It is clear that the efficiency of the motor rises gradually with vehicle speed to a maximum value of 92% at 90 km/h approximately, then decreases gradually with increasing of vehicle speed. Also, vehicle mass changing has a simple effect on motor efficiency.

$$\eta = \frac{P_{out}}{P_{in}} \quad (19)$$

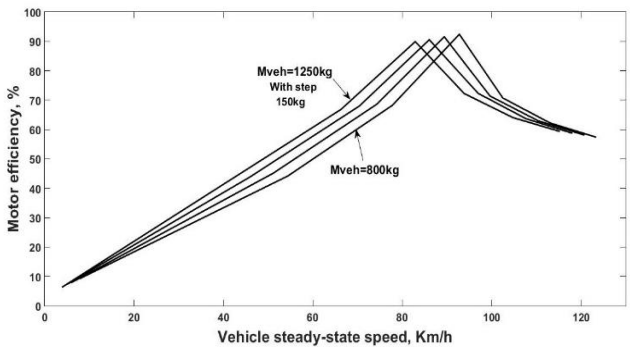

Fig.8 Motor efficiency against vehcile steady-state speed at different values of vehicle mass

Motor current is drawn versus vehicle speed from the relation (**I = T/ k_t**). At the vehicle speed of 115 km/h, the motor rated current was approximately 120 A as shown in Fig. 9.According to the catalogs of Brushless DC motors provided by M&C electric company, the motor current value 120A is sufficient to brushless DC motor with 35 KW and within the rated value [10].

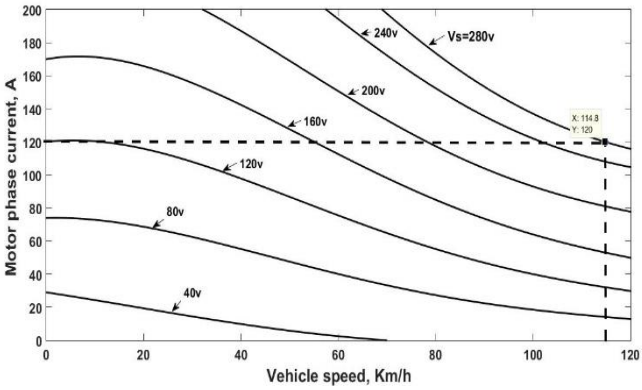

Fig. 9 Motor phase current against vehicle speed at different values of supply voltages

The maximum greadiability is an important design criterion as well as the performance measure. It is defined as the maximum grade a vehicle can climb sustaining a particular speed. It is determined by the net tractive force, Fnet or the traction force FTR of the vehicle [2,11,12]. Fnet and FTR is determined from the following equations:

$$F_{net} = F_{TR} - F_{roll} - F_{AD} \quad (20)$$

$$F_{TR} = k_m M_{veh} \frac{dv_{veh}}{dt} = \frac{T_e \eta_{tr\,m}}{r_{wh}} \quad (21)$$

The gradeability at low speeds is bigger than the gradeability at high speeds. The maximum grade that the vehicle can overcome at a given speed is expressed by [2]:

$$\tan\beta = \frac{F_{net}}{M_{veh}\,g} = \frac{F_{TR} - F_{roll} - F_{AD}}{M_{veh}\,g} \quad (22)$$

$$\text{OR } grad\% = 100\left[\frac{F_{TR}}{9.81\,M_{veh}} - c_0\right] \quad (23)$$

From the above equations, the gradeability equal to 7.3%.and that exceeds the maximum gradeability 3% [2]. So, This percentage suitable for such a vehicle to overcome the various degrees on the road.

V. CONCLUSION

A detailed study conducted at a brushless DC motor of the electric vehicle. Where characteristics of BLDCM were analyzed. The vehicle dynamics also illustrated. Then a detailed method for the sizing the major components of the electric vehicles is presented. Initially, the size of the motor has been computed. Acceleration time and acceleration distance were calculated as 9.54 seconds and 197.5 meters. That is which needed to the vehicle to accelerate from 0 to 100 Km\h. Brushless DC motor torque, output power, efficiency and current computed and plotted at different values of supply voltage versus vehicle steady-state speed using MATLAB programming. The gradeability of the vehicle was computed to be 7.3%. As a result, a 35 kW brushless DC motor is convenient to operate the vehicle at different operation conditions.

APPENDIX

a- Electrical Vehicle Parameters:
ρ= 1.225 kg/m3 , C_D=0.5 , A_F =2.6 m2 , M_{veh} = 800 kg , r_{wh}= 0.3 m , K_m= 1.08 , C_o=0.008 ,g =9.81 m/s2 , β =2°

b- Transmission Parameters:
m = 3.506

c- Data of 35 kw BLDCM:
Rated power = 35 kw, Rated speed =2500 rpm,
No. of phases =3, No. of poles =2,
Phase resistance = 0.185 Ohm, Phase inductance = 3.2 mH, Moment of inertia = $45e^{-4}$ kg.m^2,
Damping ratio = $0.28e^{-3}$ NM.

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